Windows
Command Line
Beginner's Guide



Jonathan Moeller

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THE WINDOWS COMMAND LINE BEGINNER'S GUIDE - SECOND EDITION

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Description

The Windows Command Line Beginner's Guide gives users new to the Windows command line an overview of the Command Prompt, from simple tasks to network configuration.

In the Guide, you'll learn how to:

-Manage the Command Prompt.

-Copy & paste from the Windows Command Prompt.

-Create batch files.

-Remotely manage Windows machines from the command line.

-Manage disks, partitions, and volumes.

-Set an IP address and configure other network settings.

-Set and manage NTFS and file sharing permissions.

-Customize and modify the Command Prompt.

-Create and manage file shares.

-Copy, move, and delete files and directories from the command line.

-Manage processes from the command line.

-And many other topics.

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Introduction

Welcome to “The Windows Command Line Beginner’s Guide!” If you’re reading this book, you have the opportunity to benefit from the power and flexibility of the Windows command line. Novice users can use the Windows command prompt to make repetitive tasks far quicker and easier, while advanced users can modify network settings, alter system configurations, and even use the command line to remotely control distant Windows machines.

WHAT IS THE WINDOWS COMMAND LINE?

Generally, the “Windows command line” refers to a program called the “Command Prompt”, or CMD.EXE, that has been included with every version of Windows since Windows 2000. CMD.EXE is Windows’s built-in commander interpreter. If you’re not familiar with the term, a “command interpreter” is a program that takes lines of text entered by the computer operator (i.e, you) and converts them into commands the computer can understand. A command prompt is commonly referred to as a command-line interface or a CLI.

With the Windows Command Prompt, you can open up a command-line window on your Windows system. Within Command Prompt’s window, you can use the full range of command-line utilities included with the system. You can even open multiple command-line windows, if you want to run more than one command at once.

Of course, nowadays most computers use a graphical user interface (abbreviated GUI), with a mouse pointer, windows, scroll bars, icons, and other graphical

metaphors instead of text commands. In fact, a great majority of computer users have never even used a command-line interface. Windows is famous for its graphical interface – even its name, “Windows”, refers to the graphical windows drawn on the screen of a Windows desktop or server computer. It might seem surprising that an operating system that relies upon a GUI like Windows still includes something as apparently archaic as a command-line interface.

However, Windows owes its existence to an older and simpler operating system – Microsoft Disk Operating System, MS-DOS, or as it is more commonly known, DOS. DOS originated during the early days of the personal computer revolution in the late 1970s and the early 1980s. As the personal computer market grew, IBM, a major manufacturer of high-powered mainframe computers, decided to market its own personal computer. Rather than designing specific IBM components for its personal computer, IBM decided to use off-the-shelf parts, and bought the operating system – specifically, MS-DOS, from a small Seattle company called Microsoft. IBM let Microsoft keep the rights to DOS, figuring that the real value of a personal computer lay in the hardware, not the software.

(IBM would come to regret this decision.)

As the IBM PC, and later IBM PC compatibles, took off, the PC platform faced competition from Apple’s Macintosh computer. The Macintosh relied on a GUI, which was far more user-friendly than the white text interface and cryptic error messages of the early versions of DOS. Recognizing the danger, Microsoft and IBM began joint work on a next-generation 32-bit graphical operating system called OS/2. Eventually Microsoft and IBM broke their alliance, leaving IBM with sole control over OS/2.

Microsoft pursued its own line of graphical operating systems, which it named “Windows.” The early versions of Windows were a graphical shell riding upon DOS – the user typed “win” at the command line to launch the Windows

interface. The early versions of Windows – 3.0, 3.1, and 3.11, all were graphical interfaces on top of the old DOS operating system. Even the Windows 9x line of operating systems – 95, 98, and Millennium Edition – ran on top of DOS.

However, Microsoft had created a 32-bit version of Windows called Windows NT (the "NT" stood for "new technology") to compete with IBM's OS/2. Unlike the Windows 9x line of operating systems, NT was a truly graphical operating system – the GUI did not run on top of DOS. With the release of Windows XP in 2001, Microsoft's consumer-based line of operating systems merged with the NT line of operating systems (Windows XP was technically Windows NT 5.1), and the days of DOS had come to an end.

However, Windows XP still kept the command-line interface in the form of the Command Prompt application. Windows Vista and Windows 7 & 8 followed suit, as did Microsoft's line of server operating systems – Windows Server 2003, Windows Server 2003 R2, Windows Server 2008, Windows Server 2008 R2, and Windows Server 2012. DOS is history, but thanks to the Command Prompt application, you can still use the old DOS commands on a modern Windows system.

WHY LEARN THE COMMAND LINE?

You might wonder why you would want to learn the command line, since Windows comes with a perfectly fine graphical user interface. Just because Windows comes with the command line doesn't mean you have to run it, after all. And what possible use could you get out of it?

For one, the command line permits you to perform numerous repetitive and tedious computing tasks far more quickly. For example, let's say you need to copy a large number of JPEG image files out of your Pictures folder and onto

your flash drive (which Windows has assigned the letter J:). Your JPEGS are named for the month and day they were taken – August01.JPEG, March01.JPEG, and so forth. Specifically, let's say that you need to copy every JPEG taken in the month of August to your flash drive.

To copy the files, you could drag-and-drop every single one of the files, or CTRL-click each of the files, and then drag them to your flash drive. Either way, it would take a great deal of bother and hassle.

Or you could simply type this command at the Command Prompt:

```
COPY c:\username\pictures\August*.JPEG J:\
```

This command will copy every JPEG file beginning with "August" in your Pictures folder to your flash drive. Needless to say, this is vastly more efficient than dragging and dropping, copying & pasting, or CTRL-clicking from the GUI.

Information technology professionals, such as help desk technicians and network administrators, have an even greater motivation to learn the command line. Everything in Windows can be done through the GUI. However, many tasks can be performed far more efficiently at the Command Prompt. In particular, it is easier to do numerous network tasks from the command line than through the GUI. For example, finding your IP address on a Windows 7 desktop system takes five different mouse clicks and four separate dialog boxes. Finding your computer's IP address from the Command Prompt takes one brief command.

And on a Windows system, many network diagnostic tools are available only

through the command line. If your system loses its network connection, the graphical tools aren't terribly helpful. The command line utilities are far more efficient at discovering the root of the problem, if you know how to use them properly.

Finally, the command line offers the advantage of automation. An IT professional will often find himself dealing with repetitive and tedious tasks. Using the Command Prompt, is possible to create simple scripts called "batch files" to deal with these boring jobs. Instead of typing the same sequence of commands over and over again, or wading through graphical menu after graphical menu, a well-written batch file can handle these tasks with a single typed command.

THE PURPOSE OF THIS BOOK

The purpose of this book is to provide a basic introduction to using the Windows Command Prompt. It's not intended as an exhaustive or comprehensive overview, but as an introduction to the topic - enough to get you started, and comfortable enough to start experimenting on your own. (Feel free to jump around if a particular topic interests you more than the others.) We can divide this book into three parts:

-Part I: Command line basics and file management basics.

-Part II: Networking.

-Part III: Advanced tasks - disk management, remote management, batch files, and others.

WHY I WROTE THIS BOOK

I began writing about computers and technology almost by accident. I started my writing career as a writer of fantasy fiction (my sword-and-sorcery novels are now available in all major eBook formats), and like every good fantasy writer, I had a blog that tended to pull down only thirty or forty hits a month. One day I happened to write about a problem I had with a file server, and the next day I was surprised to see nearly sixty hits come in via Google searches for that same problem. From there it was a short step to blogging regularly about technology (and turning a profit via web ads), and then when the eBook revolution struck, to writing short eBooks on technical topics.

I decided to write this book because an entire generation of computer users, those born after 1990 or so, have no experience of using a command-line interface. Many high school and college students only began using computers after GUIs became dominant, and therefore have no idea how to use the Command Prompt, or are even aware that it exists. It is my hope that this book will help anyone to learn to use the Windows command-line interface, and show them ways to make their computer more efficient and useful.

A NOTE ABOUT WINDOWS VERSIONS

All the commands discussed in this book will work on Windows 7, Windows 8, Windows 8.1, Windows 10, Windows Server 2008, Windows Server 2008 R2, Windows Server 2012, and Windows Server 2012 R2. Most of them will work on Windows Vista, Windows Server 2003 R2, and Windows Server 2003. Many of the commands will work on older versions of Windows, but the older the version of Windows, the less likely the commands will work. In particular, many of the network commands used in the modern versions of Windows will not work on the older Windows 9x family of operating systems.

ERRATA

I have done my best to make sure that all the information in this book is accurate and timely, and tested every command and procedure described in the following chapters. However, I am only mortal, and undoubtedly I have made mistakes. If you notice any errors, you can email me at jmcontact @ jonathanmoeller.com to let me know. The advantage of eBooks over paper books is that eBooks are vastly easier to update and revise, and I can quickly introduce a revised and updated version to correct any mistakes. (Another advantage of ebooks over paper books is that you can have it open on your computer screen as you try out the commands, rather than having to look down at a paper book on your desk.)

NOTES ON THE SECOND EDITION

This book has sold far better than I anticipated. That, combined with changes in Windows 8. Windows 8.1, and Windows 10 made it necessary to create a second edition. In this edition, I have added sections dealing with the Command Prompt in these versions of Windows. I have also added a new chapter describing how to perform task management from the command line itself, and a new section in Chapter 1 describing how to change the size of the Command Prompt window.

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Chapter 1 - Command Line Basics

By default, Windows boots into a graphical interface. In fact, if Windows boots into something other than the default graphical interface, you can safely assume that something has gone wrong with your system (or that you have booted into one of the versions of Safe Mode). How, then, to launch the Command Prompt from the GUI?

In this chapter, we'll discuss how to launch the Command Prompt, and a few basics about using the Windows command line. We'll also talk about command history, how to copy and paste, the case sensitivity of the Command Prompt, and a few other useful topics.

LAUNCHING THE COMMAND LINE

You can launch the Command Prompt like any other Windows application. To do so, go to the Start Menu, and then to the Accessories folder. Inside the Accessories folder, you'll see the icon for the Command Prompt – it will look like a little black square with white text in it. Click on that icon, and the Command Prompt window will launch.

You can also launch the Command Prompt using the Start Menu Search feature that has been included with every version of Windows since Vista launched in late 2006. When you click on the Start Menu button, you'll see a Search field at the bottom of the left-hand pane in the Start Menu. In that field, type "command", and the Command Prompt should appear as the top hit. To find it even faster, just type "cmd" (the Command Prompt's real name is CMD.EXE, after all), and again the Command Prompt should be the top hit.

There is an even faster method of launching the Command Prompt. Most computers produced in the last ten years include the “Windows key”, which is usually between the CTRL and ALT keys on the left side of the keyboard (desktop USB keyboards generally include a second Windows key between the right CTRL and ALT keys). If you hit the Windows key and the R key simultaneously, this will bring up the Run box. If you type the name of a program or file into the Run box, Windows will launch or open it for you. You can launch the Command prompt by simply typing “cmd” into the Run window and hitting ENTER or clicking OK.

LAUNCHING THE COMMAND LINE FROM WINDOWS 8, 8.1, & 10

If you have used Windows 8, you know that the new user interface (commonly called the “Modern Style UI”) in Windows 8 is very different from the traditional Windows environment. Fortunately, launching the Command Prompt from Windows 8 is quite simple, and Windows 8 offers options for launching the Command Prompt that previous versions of Windows do not possess.

To launch the Command Prompt in Windows 8, tap the WINDOWS key to summon the Start Screen. Once the Start Screen appears, you can use its integrated search function to find the Command Prompt by typing “cmd.” (There’s no actual “search box” – just start typing at the Start Screen, and Windows 8 will begin searching for you.) Once you type “cmd”, the tile for Command Prompt will appear. Click on it, and you will be taken from the Start Screen to the Desktop, which will have an open Command Prompt window.

Windows 8 and up also include File Explorer, an improved version of the Windows Explorer shell. File Explorer offers the ability to launch a Command Prompt window within the folder you are currently viewing. To do so, click on the File tab in the upper left-hand corner of the File Explorer window, and then

select “Open Command Prompt” from the menu. You will have the option to choose between launching a Command Prompt window as a standard user or as a administrative user. (We’ll discuss that, and User Account Control, more later in this chapter.)

For more information about Windows 8 in general, I suggest my book The Windows 8 Beginner's Guide (http://www.jonathanmoeller.com/writer/?page_id=2797).

RUNNING THE COMMAND LINE AS AN ADMINISTRATOR

From time to time you might get in an error message in the Command Prompt that “this operation requires elevation” or that “this command must be run as an administrator.” What does that mean?

Windows Vista, Windows 7, Windows Server 2008 / 2008 R2, and Windows Server 2012 and up use a security technology called “User Account Control.” User Account Control (UAC) is a method of locking down user accounts in order to protect Windows systems from viruses, Trojans, and other forms of malware. Basically, a Windows system has two kinds of user accounts – administrative user accounts, which can install programs and alter system settings, and standard user accounts, which can run programs, but can’t install software or alter settings. In Windows XP, most people used local administrator accounts, which made it easy for malware to install itself undetected in the background.

UAC provides a level of protection against that kind of attack. Under UAC, all accounts, even administrator accounts, run as normal accounts. When you do something that requires administrative privileges, such as installing a program, or trying to change a system setting, the screen freezes and the UAC prompt

appears. For administrative users, the UAC prompt reads “Do you want to allow the following program to make changes to this computer?” followed by buttons for Yes or No. Standard users are prompted to enter an administrative username and password to continue.

Because of this, Command Prompt usually opens with the permissions of a standard user, even when you launch it from an administrator's account. This means you won't be able to use Command Prompt to launch any commands that alter the system or install programs. To do so, you will need to run Command Prompt as an administrator.

Fortunately, running any program as an administrator within Windows is easy. Simply right-click on the icon for Command Prompt and select "Run as administrator" from the context menu that appears (it will have the multicolored shield icon of the UAC prompt next to it). If you're running within an administrator account, you need only click Yes to continue; if you're using a standard user account, you'll need to enter an administrator's username and password. Once Command Prompt opens as an administrator, the program's title bar will change from "Command Prompt" to "Administrator", and you can now launch commands that will affect system settings.

MAKING THE COMMAND PROMPT WINDOW LARGER

By default, the Command Prompt window is not very large. Even if you maximize it, the window will not take up more than a third of your screen or so. If you want a larger window, right-click on the Command Prompt window’s title bar, and select Properties. When the Properties window opens up, click on the Layout Tab.

The Layout tab lets you control the size of your Command Prompt window. The

easiest way to do so is to adjust the numbers in the Window Size category. By adjusting the Height and the Width numbers, you can make the Command Prompt window as small or as large as you like. Once you have finished, click on the OK button, and the size of the Command Prompt window will change to your specifications.

(Personally, I find that a small window is more useful, since that allows you to read directions off a web page or an ebook while you work.)

TAB COMPLETION

Back in the old days of DOS, filenames and directory names were limited to only eight characters, with no spaces, and another three characters to denote the file extension (used by the operating system to determine what program to use to open the file - a TXT extension means that the file is a text file, for example). These were called 8.3 filenames, and as you can imagine, the limitation led to some creative and cryptic naming of documents. A file might have a name like "THSREV2.DOC", and it would take a creative thinker indeed to realize that the file name refers to the second revision of the author's senior thesis.

All modern versions of Windows support "long file names", which means you can have file names up to 255 characters long. And the file names can even include spaces! So instead of "THSREV2.DOC", the author could name his file "Second Revision of Senior Thesis.DOC" with room to spare.

However, short file names have one advantage - they're easier to type at the command line. Typing "THSREV2.DOC" as the Command Prompt is easy, but typing "Second Revision of Senior Thesis.DOC" takes rather longer.

This is where tab completion comes in handy.

Type the first part of a file name, hit TAB, and Command Prompt will take its best stab at filling out the rest of the file's name for you. For instance, let's say you wanted to use the DEL command to delete a file named test.txt located at C:\users\camalas\documents\test.txt. You could type out the entire file name and path.

Or you could type the command and the first part of the path:

DEL C:\us

Hit the TAB key, and the command will look like this:

DEL C:\Users

Add an additional backlash and the letter C, hit the TAB key, and Command Prompt will fill out the next step in the path:

DEL C:\Users\camalas

Using tab completion, you can fill out the entire file location bit by bit, which is far quicker than typing out the entire path.